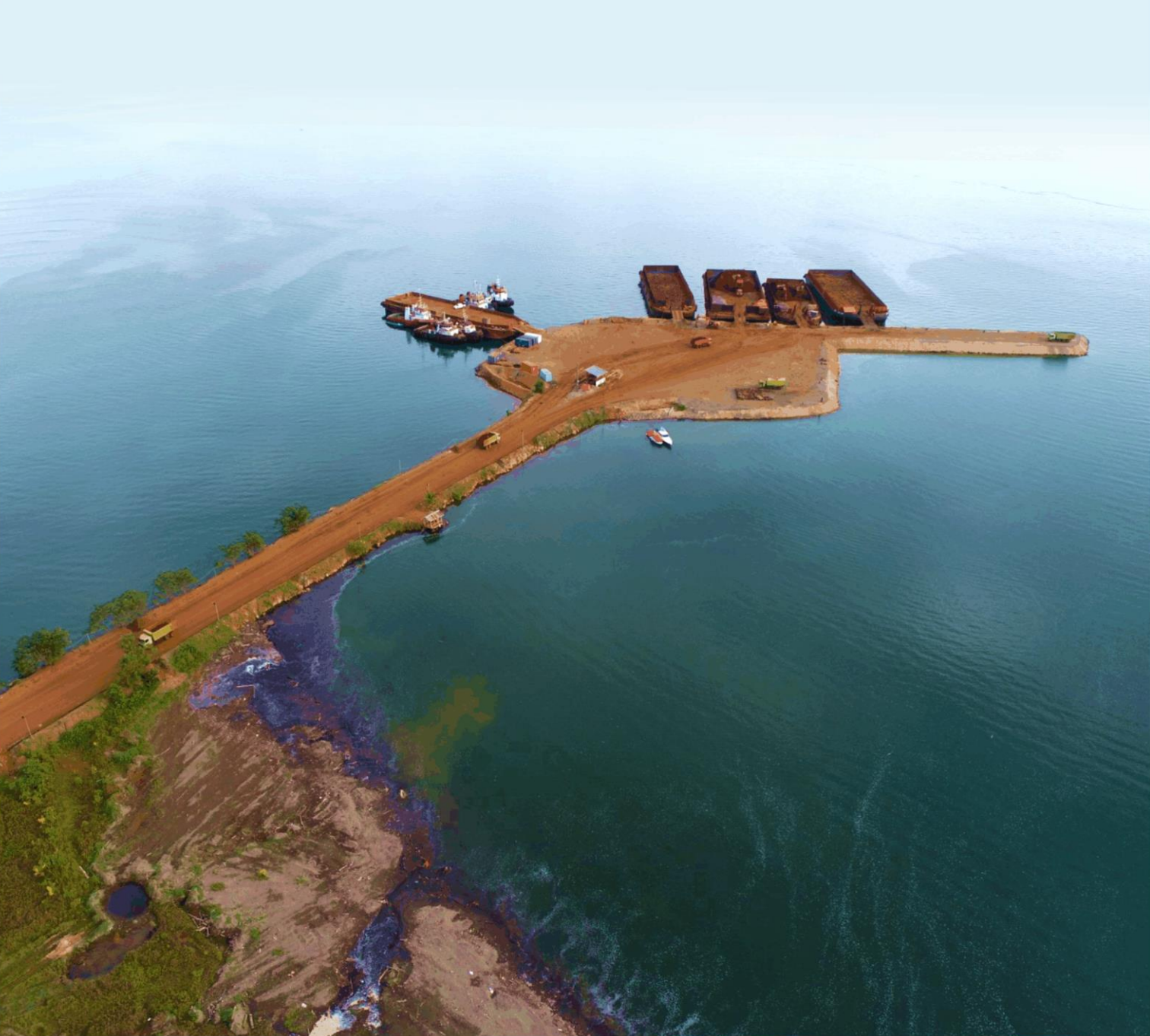
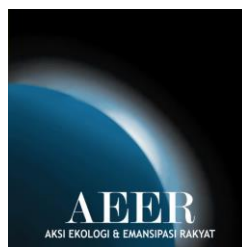




THE ROLE OF ESG IN THE IMPROVEMENT OF NICKEL INDUSTRY FOR ELECTRIC VEHICLES

The Role of ESG in The Improvement of Nickel Industry for Electric Vehicles



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viii + 26 pages. (20 cm x 28 cm)

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Front & Back Cover Photo: Christ Belseran / Mongabay Indonesia

Published by:

Perkumpulan Aksi Ekologi dan Emansipasi Rakyat (AEER)

<http://aeer.or.id>

Talavera Office Park, 28th floor

Jl. TB Simatupang Kav 22-26, Jakarta 12430

June 2024

List of contents

Executive Summary v

Recommendations vii

1. Indonesian Nickel & Global Electric Vehicle (EV) Industry 1

2. Global Nickel Supply Dynamics 9

3. Nickel Supply Chain from Indonesia to Tesla, BYD, & Volkswagen 13

4. Relevance of Environmental, Social, and Governance (ESG) Assessment
and PROPER Rating in the Nickel Industry for EV Batteries 17

5. Identification of ESG & PROPER Assessment of Nickel Production Company
for EV Battery 21



Sumber Gambar : automacha.com

Executive Summary

Global climate change is contributed by the increasing carbon emission from human activities since the 1800s, mainly by burning fossil fuels. In 2019, the transportation sector contributed as many as 8.7 GtCO₂-eq or equal to 23% of the energy-related global emissions, where 70% is contributed by land transportation. One of the decarbonization efforts is the transition of the transportation sector to electric vehicles. Electric vehicle (EV) is a vehicle powered by an electric motor that receives its energy from a battery and can store energy through power charges. One of the essential battery components is the nickel mineral. The transition to the low emission vehicle increases the demand for EVs which is in line with the increase of nickel as a battery component.

Indonesia is the biggest nickel producer country in the world with reserve amounting to 52% of the world nickel reserve. Seeing the potential of global nickel demand, the Government of Indonesia have been trying to leverage this opportunity to improve the national economy. In the attempt, the Government of Indonesia has banned nickel ore export and has boosted smelters development to not only produce semi-finished nickel, but also to produce battery nickel. Huayou Indonesia, a producer of battery-component nickel in Indonesia, supplies products for the Contemporary Amperex International (CATL) that is located in China. CATL is the battery supplier for BYD, Tesla, and Volkswagen, the top three EV manufacturers.

Nickel battery production should pay attention to the social-environmental aspects to benefit all parties and to avoid negative social-environmental impacts. Internationally, ESG assessment exists to evaluate the company's business performance and sustainability in the environment, social, and governance aspects. At the national level, Indonesia has the PROPER assessment under the Ministry of Environment and Forestry that assesses the environmental management (of wastewater, gas emission, and toxic & hazardous waste). Companies that are publicly listed on the Indonesian Stock Exchange are required to perform and report their ESG, whereas those not listed do not have the same responsibility. For these unobligated companies, a PROPER rating could be used to assess the responsibility in the environmental sustainability aspect.

This study discovered nickel production companies publicly listed on the Indonesia Stock Exchange (IDX) are not yet transparent and open on the risks of their activity to the environment, social, and governance. The public does not even have information access to the scores of each metric in the ESG assessment. In addition, what the public experiences, especially the local community, are often not congruent with the company report contents. In addition, the nickel smelters for battery manufacturers in Indonesia (Huayue NC, QMB NEM, and Halmahera PL) are not even listed on the IDX, therefore they do not have the obligation to report their business practice under the framework and standards of the ESG.

For the PROPER assessments, 7 out of 15 companies in the Indonesia Morowali Industrial Park (IMIP) area received red ratings, which means the environmental management is not yet complying with the requirements stipulated in the law. The important note is that there have not been any attempts to improve the environmental aspect by the company, indicated by two consecutive years of red rating. Whereas companies with a PROPER red rating receive administrative sanction in the form of instruction to improve and the obligation to attend training for at least three (3) months after the PROPER rating is determined.

Moreover, the energy demand for nickel production process in Indonesia is still supplied by captive coal power plants in a big portion of 8,817 MW, or equivalent to 64.2% of the total captive coal power plant energy. In the IMIP industrial area, the electric power requirement is supplied by Sulawesi Mining Power Station, which is a captive coal power plant. The power plant is jointly owned by PT Sulawesi Mining Investment, Indonesia Guang Ching Nickel and Stainless Steel Industry, Indonesia Tsingshan Stainless Steel, and Indonesia Morowali Power. The power plant consists of 9 already operating units with a total capacity of 2,080 MW (or 2.08 GW). As an illustration, 1 ton of nickel matte produced from nickel pig iron (NPI) conversion will emit 59 tons of CO₂, production of 1-ton mixed hydroxide precipitate (MHP) with High Pressure Acid Leaching (HPAL) technology emits 19 tons of CO₂. This activity is not aligned with the global Net Zero Emission ideals adapted by Indonesia as *Enhanced National Determined Contribution* (ENDC) for unconditional emissions reduction of 31.89% equivalent to 915 MTon CO₂-eq and the conditional emissions reduction with international assistance of 43.2% equivalent to 1,240 MTon CO₂-eq by 2030.

The electric vehicle production increase with the purpose to reduce global carbon emissions becomes incongruous if the nickel used is produced with energy from coal and through practices that disregard ESG. EV manufacturers as the receiver of nickel for battery has an influence to the market and nickel standard by stopping the "dirty" nickel supply for their EV production.

Recommendations

1. Tesla, VW, and BYD are demanded to stop supplying their manufacturers with batteries and battery components of nickel sourced from mining and processing industries that perform poorly in social and ecological terms and do not practice sustainability aspects in their production. This could be observed in the company performance evaluation, both at the international level in the form of ESG score or the Indonesian national level in the form of PROPER rating.
2. Tesla, VW, and BYD are demanded to stop supplying their manufacturers with batteries and battery components of nickel produced using coal energy sources. This would be highly contradictory to the fact that EVs exist for the purpose of decarbonizing global emissions.
3. Transparent sustainability practices of nickel supply companies in Indonesia are needed. This could be achieved in the form of company annual reports, corporate sustainability reports, and publication of ESG practices in corporate communication media that can be easily accessed by the public.
4. Initiatives are needed to reduce dependence on primary nickel from mines. Efforts to improve nickel circularity can be made through nickel battery recycling¹. Through this approach, nickel is not only the key to the development of EV technology, but also an important part of the bigger solution in the global energy transition toward more green and sustainable ways.
5. EV manufacturers need to switch to more environmentally friendly battery technologies in their production process.

¹ Powershift. (2024). Reducing metal consumption: Practical suggestions for the raw materials transition. <https://power-shift.de/wp-content/uploads/2024/04/Reducing-metal-consumption.pdf>

Picture Source : policyoptions.irpp.org



1. Indonesian Nickel & Global Electric Vehicle (EV) Industry

Climate change is a long-term change in the patterns of temperatures and climates. Such changes in patterns can occur naturally by solar activity or volcanic eruptions. However, since the 1800s, human activities become the main driver of climate change, mainly by the burning of fossil fuels; coal, oil, and gas. Burning fossil fuel produces greenhouse gas (GHG) emissions that trap heat from sunlight and increase temperatures.²

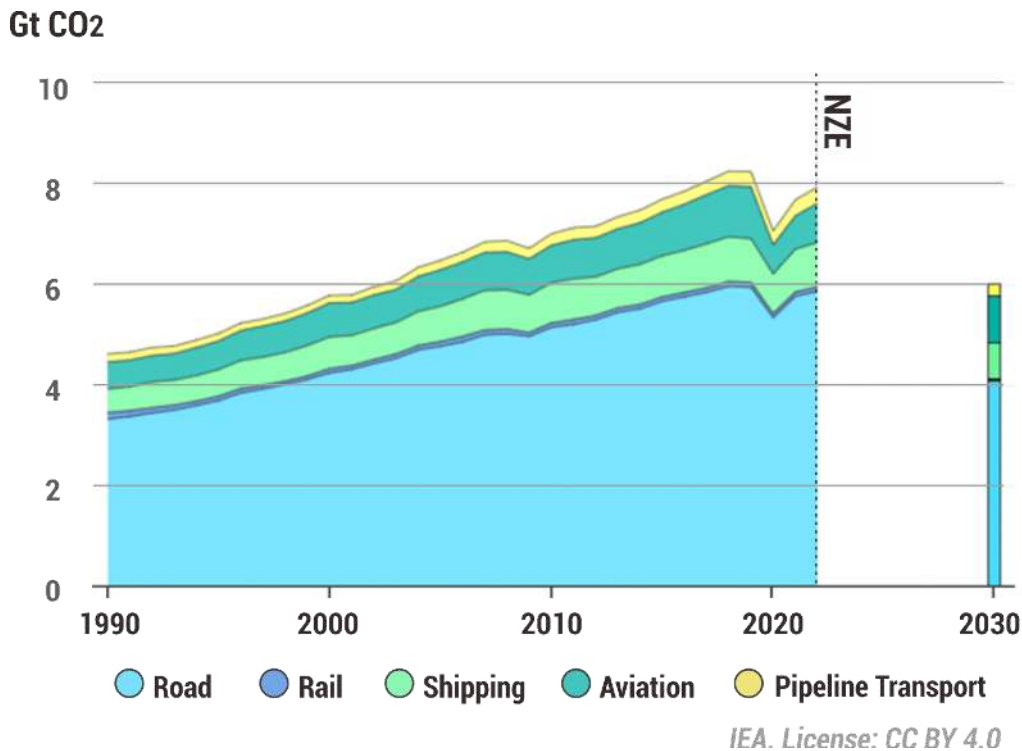
In 2019, GHG direct emission from the transportation sector was 8.7 GtCO₂-eq (an increase from 5.0 GtCO₂-eq in 1990) and contributed to 23% of the global energy-related CO₂ emissions. 70% of direct emissions from the transportation sector come from land vehicles, while 1%, 11%, and 12% are from the railways, shipping, and flight respectively.³ The trend of transportation sector emissions showed a decline in 2020, which showed a return incline in 2021 & 2022 with road transportation emissions recorded at 5.87 GtCO₂-eq in 2022.



Picture Source : faz / carfromjapan.com

² <https://www.un.org/en/climatechange/what-is-climate-change>

³ Jaramillo, P., S. Kahn Ribeiro, P. Newman, S. Dhar, O.E. Diemuodeke, T. Kajino, D.S. Lee, S.B. Nugroho, X. Ou, A. Hammer Strømman, J. Whitehead, 2022: Transport. In IPCC, 2022: Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [P.R. Shukla, J. Skea, R. Slade, A. Al Khourdajie, R. van Diemen, D. McCollum, M. Pathak, S. Some, P. Vyas, R. Fradera, M. Belkacemi, A. Hasija, G. Lisboa, S. Luz, J. Malley, (eds.)]. Cambridge University Press, Cambridge, UK and New York, NY, USA. doi: 10.1017/9781009157926.012



Among the attempts to reduce emissions from the transportation sector is transitioning from fossil fuel vehicles to electric vehicles (EVs). The claim is supported with EV direct emission is smaller at 60 g CO₂/km compared to conventional vehicles at 130 g CO₂/km.⁵

Nickel has an important role in the EV industry, since it is the main material for lithium-ion batteries used for electric vehicles. The global transformation towards a more environment-friendly vehicle has put nickel as the critical component in this journey. Entering the era of electric vehicles, the challenge is to create batteries that are not only efficient and strong, but also affordable and sustainable. This issue is increasingly urgent as demand for electric vehicles continues to rise. Nickel content in lithium-ion batteries is key in answering these needs. Nickel is able to increase energy storing capacity and efficiency, and it is a cheap resource to reduce the production cost of electric vehicle batteries.⁶

⁴ IEA (2023), Greenhouse Gas Emissions from Energy (<https://www.iea.org/data-and-statistics/data-product/greenhouse-gas-emissions-from-energy>).

⁵ <https://www.seai.ie/technologies/electric-vehicles/why-drive-electric/the-environment/>

⁶ <https://nickelinstitute.org/en/about-nickel-and-its-applications/nickel-in-batteries/#:~:text=The%20major%20advantage%20of%20using,capacity%20at%20a%20lower%20cost.>

Indonesia is the biggest nickel producer in the world. According to the Ministry of Energy and Mineral Resources, nickel reserves in Indonesia are 52% of the world's nickel reserves or equivalent to 72 million tons of nickel⁷. In 2022, Indonesia was recorded as the biggest nickel producer in the world with 1.6 million metric tons of production and 21 million metric tons of nickel reserve. Indonesia holds an important role in the global supply chain for nickel and its derivatives, which are very vital for EV battery manufacturing.

DISTRIBUTION OF INDONESIAN NICKEL ORE RESERVES

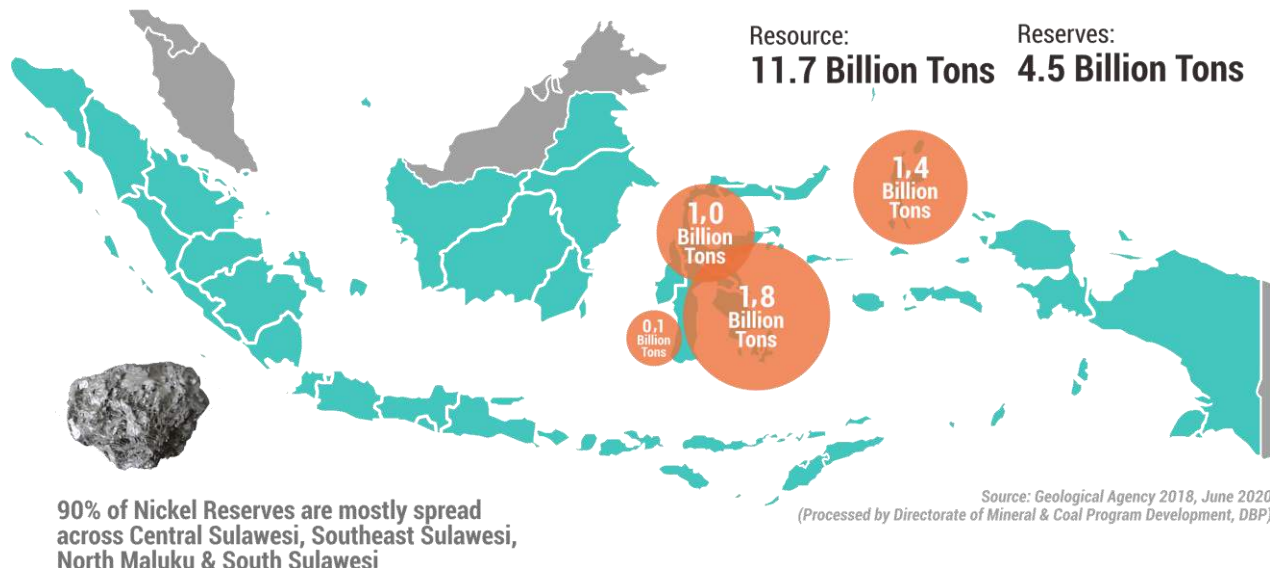


Figure 2. Nickel reserve distribution in Indonesia⁸

Indonesia has a policy that bans nickel ore export, it was first implemented in 2014. The policy is part of the national economy strategy with its main goal of increasing sovereignty and control the use of natural resources, increasing public welfare, and strengthening the country's economy. The policy is designed to strengthen Indonesia's bargaining position in global trade, demonstrating the independence and courage in decision-making that will provide long-term benefits for the country. Indonesia's decision to ban nickel ore export is the continuation of the industry policy to produce materials and downstream products in the nickel and EV battery supply chain. This step is also in line with the effort to develop the industry of value-added nickel products for the stainless-steel supply chain.

⁷ Ministry of Energy and Mineral Resources Indonesia. (2020). Booklet Tambang Nikel.

⁸ *ibid*

With the biggest nickel reserves in the world, which are mainly class-2 nickel, Indonesia's resources are not suitable for EV batteries. To overcome this gap, Indonesia is developing nickel technology from class-2 to class-1, including the high-pressure acid leaching (HPAL) process. However, the HPAL process causes environmental concerns because of its water consumption, its production of caustic tailing (chemically destructive), and its high energy usage.

In terms of regulation, the amendment of Law No. 3 of 2020 on Mineral and Coal Mining stipulated the policy that mineral and coal management should be controlled by the central government to provide real added value to the national economy. This law enables the government to establish mineral and/or coal policies that prioritize domestic interests, which can be realized through production and export controls.

In addition, through the Ministry of Energy and Mineral Resources Regulation No. 11 of 2019 on Second Amendment to the Minister of Energy and Mineral Resources Regulation No. 25 of 2018 on Mineral and Coal Mining Business, Indonesia optimizes the potential of the nickel industry through nickel downstream strategy. By emphasizing domestic nickel processing and developing nickel smelter facilities, Indonesia attempts to maximize the benefit of its abundant natural resources. By March 2024, 44 nickel smelters are operating in Indonesia, which will increase with an additional 19 smelters currently in construction and 7 smelters currently in the feasibility study phase.⁹

However, the increasing number of smelters in Indonesia poses a new threat to the global Net Zero Emission goal. Because the energy demand for the manufacturing processes is still supplied by captive coal power plants. Burning coal releases CO₂, fine particles (PM_{2.5}), also SO₂ and NO_x gasses to the air. CO₂ is a climate change driver, PM_{2.5} is detrimental to human health when inhaled, and NO_x contributes to climate change and environmental acidification and is toxic.¹⁰ During 2015-2019, 44% of global emissions came from coal burning.¹¹ Burning coal for energy purposes emits 740-1,689 g CO₂e/kWh.¹²

⁹ <https://www.bloombergtechnoz.com/detail-news/32862/ri-banjir-smelter-nikel-44-beroperasi-per-maret-19-menyusul>

¹⁰ Oberschelp, C., Pfister, S., Raptis, C.E. *et al.* (2019). Global emission hotspots of coal power generation. *Nat Sustain* 2, 113–121. <https://doi.org/10.1038/s41893-019-0221-6>

¹¹ Clarke, L., Y.-M. Wei, A. De La Vega Navarro, A. Garg, A.N. Hahmann, S. Khennas, I.M.L. Azevedo, A. Löschel, A.K. Singh, L. Steg, G. Strbac, K. Wada, 2022: Energy Systems. In IPCC, 2022: Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [P.R. Shukla, J. Skea, R. Slade, A. Al Khourdajie, R. van Diemen, D. McCollum, M. Pathak, S. Some, P. Vyas, R. Fradera, M. Belkacemi, A. Hasija, G. Lisboa, S. Luz, J. Malley, (eds.)]. Cambridge University Press, Cambridge, UK and New York, NY, USA. doi: 10.1017/9781009157926.008.

¹² IPCC. (2014). AR5 Synthesis Report

It can be seen in Figure 3 that nickel production in Indonesia takes up the largest portion, at 64.2% of the total energy provided by captive coal-fired power plants. Based on the Global Energy Monitor data¹³, the electrical energy needs of the IMIP area are provided by the Sulawesi Mining Power Station, which is a captive coal power plant. The power plant is jointly owned by PT Sulawesi Mining Investment, Indonesia Guang Ching Nickel and Stainless Steel Industry, Indonesia Tsingshan Stainless Steel, and Indonesia: Morowali Power. The power plant consists of 9 already operating units with a total capacity of 2,080 MW (or 2.08 GW). As an illustration, 1 ton of nickel matte produced from nickel pig iron (NPI) conversion will emit 59 tons of CO₂, production of 1-ton mixed hydroxide precipitate (MHP) with HPAL technology emits 19 tons of CO₂¹⁴. And yet, in the Enhanced Nationally Determined Contribution, Indonesia targets to reduce emissions with its own efforts (unconditional) by 31.89% and with international assistance (conditional) by 43.2% by 2030¹⁵. The use of electric vehicles, which aims to reduce global carbon emissions, becomes inconsistent if the nickel used is produced with energy from coal.

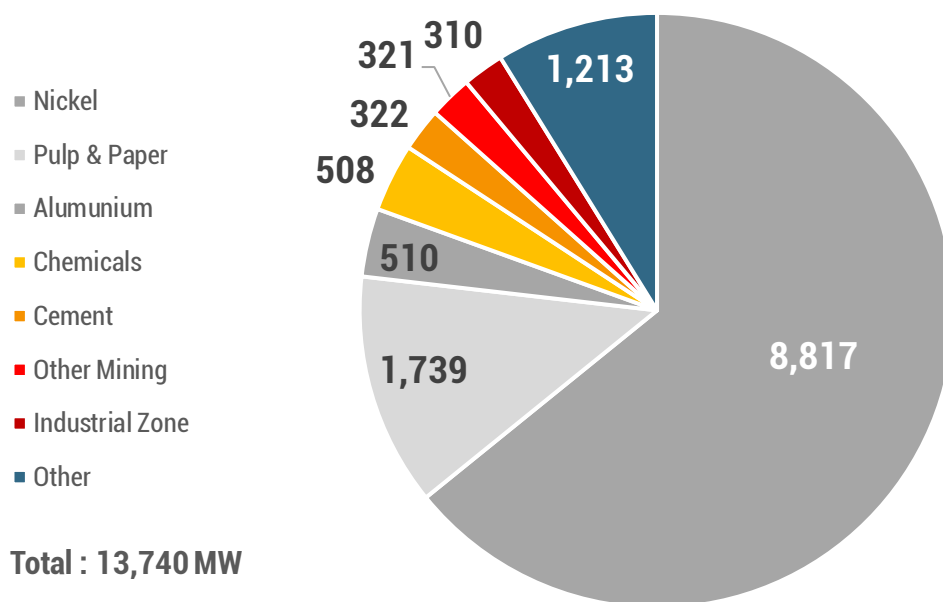


Figure 3. Supply portions of captive coal power plants in Indonesia¹⁶

¹³ <https://globalenergymonitor.org/projects/global-coal-plant-tracker/tracker/>

¹⁴ IEA (2021), GHG emissions intensity for class 1 nickel by resource type and processing route, IEA, Paris <https://www.iea.org/data-and-statistics/charts/ghg-emissions-intensity-for-class-1-nickel-by-resource-type-and-processing-route>, Licence: CC BY 4.0

¹⁵ Indonesia. (2022). Enhanced Nationally Determined Contribution Republic of Indonesia

¹⁶ Asian Development Bank. (2023). Captive Power Landscape Assessment for the Energy Transition in Indonesia. Final Report - Executive Summary.

By 2023, global EV sales were expected to reach 13.8 million units, increasing from 12% in 2022 to 16.2%. Overall, the global EV market is predicted to reach 27.1% in 2024, or as many as 17.5 million units.

Regionally, China leads the market with a 55.5% market share, which is expected to deliver 7.6 million EVs by 2023. In addition, Europe and North America are also significant markets, with Europe reaching sales of 3.2 million units by 2023, followed by North America at 1.8 million units. The trend of EV global sales can be seen in Figure 4.

The development is supported by several factors, including technological advancement in battery components and high voltage components. These components are important for the EV functions and are the biggest part of the production cost. However, over time, the cost of these components will decrease, which will drive market growth.

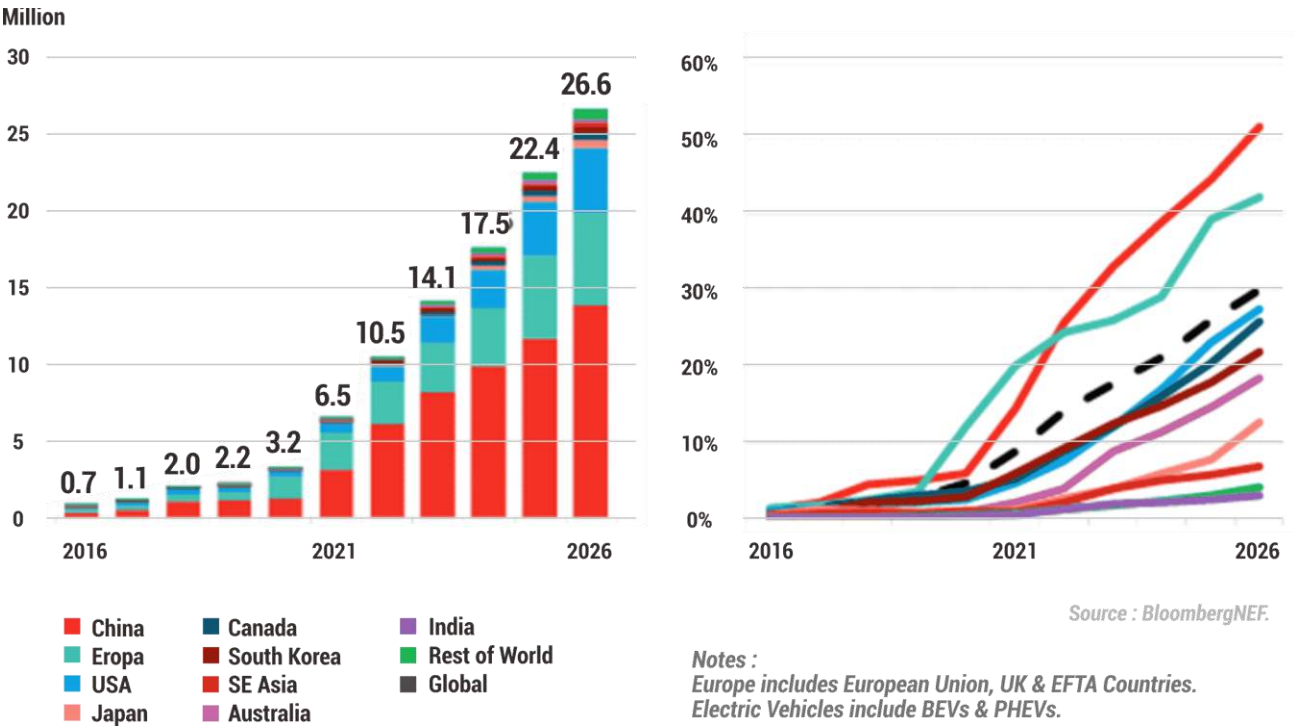


Figure 4. Global EV sales trend¹⁷

Currently, the EV industry has three prominent manufacturers globally: Tesla, Volkswagen (VW), dan BYD. All three have demonstrated significant innovation in EV technology and consistently lead the market in electric vehicle production and sales.

¹⁷ BloombergNEF. (2023). Electric Vehicle Outlook.

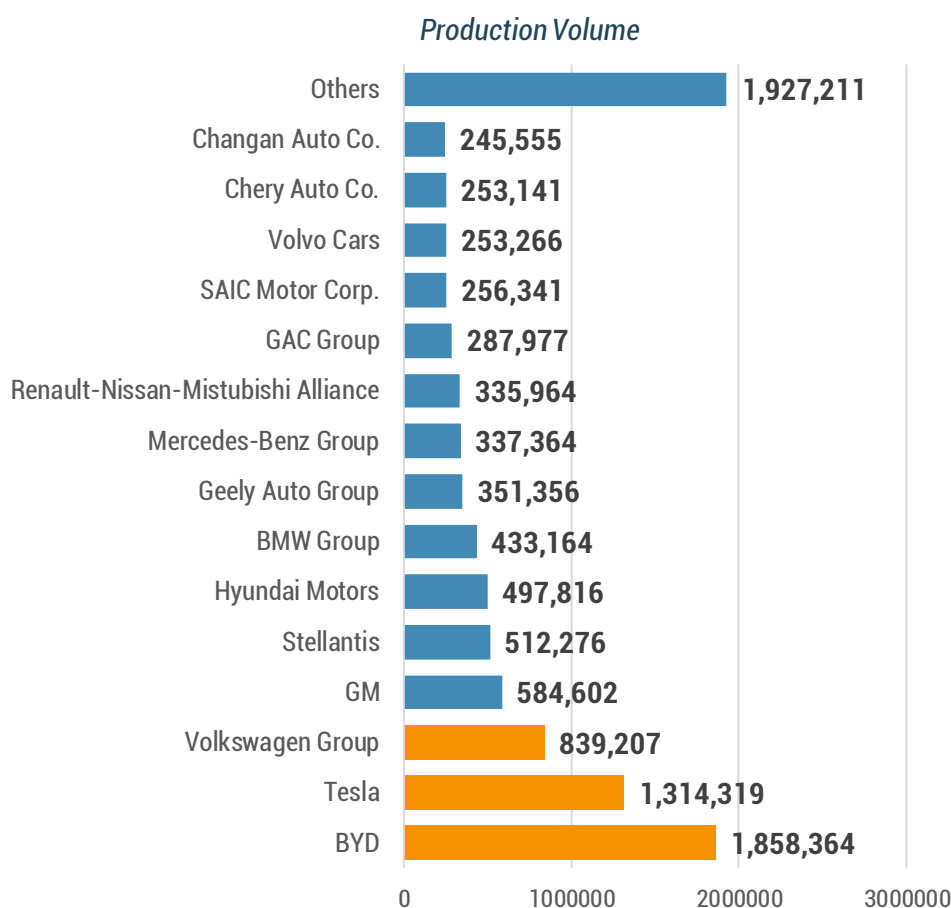


Figure 5. EV production by leading global manufacturers¹⁸

Data for 2022 shows BYD leading the way with 1.86 million EVs, followed by Tesla with 1.3 million EVs, and in third place is VW Group with 839,207 EVs. If on average one EV unit requires 25.3 kg¹⁹ of nickel, the total nickel demand for BYD, Tesla, and Volkswagen production in 2022 is 101,500.817 tons. The production process of 1 kg of nickel metal emits 13 kg of CO₂, with approximately 60% being direct emissions, 15% emissions from electricity consumption, and 25% emissions from chemical processes used during the production process.²⁰ In addition, 100 kg of laterite ore source is required to produce 1 kg of nickel. When processed with acid, it will result in 99 kg of acid waste with iron content.²¹

¹⁸ <https://www.statista.com/statistics/271013/worldwide-leading-electric-car-manufacturers-based-on-vehicle-production/>

¹⁹ <https://www.adamasintel.com/increase-in-nickel-deployed-in-evs-by-sales-weighted-average/#:~:text=Amount%20of%20nickel%20in%20average,many%20entry%2Dlevel%20new%20models.>

²⁰ Nickel Institute. (2020). Life Cycle Data Question & Answers.

²¹ <https://source.benchmarkminerals.com/article/nickel-the-often-forgotten-battery-metal>



Picture Source: australiansolarquotes.com.au

2. Global Nickel Supply Dynamics

Global trends in nickel supply and demand can be obtained through projections of nickel demand and supply up to 2030 in Figure 6, sourced from Bloomberg NEF. The graph shows nickel supply demand is classified into four categories: EV demand, electronics/stationary storage demand, non-battery demand, and non-battery flexible demand. We can see that EV demand is increasing most rapidly. Assuming that the increase in EV demand is directly proportional to the production of lithium-ion batteries, the steep trend in EV demand may indicate a significant impact of the EV market on the demand for nickel, which is the main constituent component of batteries.

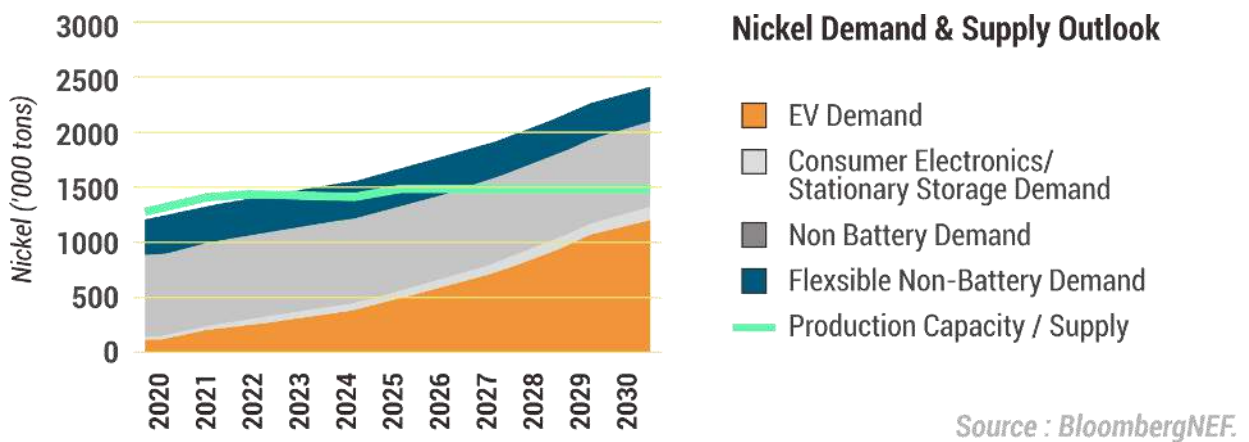
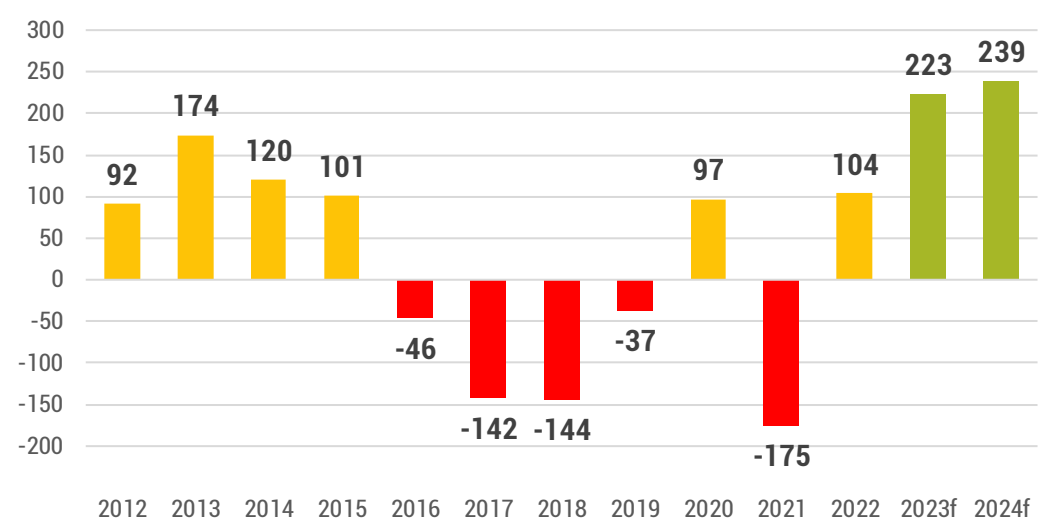


Figure 6. Projection of nickel demand 2030 and sectors of utilization²²

²² <https://stockhead.com.au/resources/asx-juniors-are-feeding-off-nearology-to-was-nickel-producing-hotspots/>

Along with global efforts to decarbonize transportation systems, EVs will require large amounts of high-purity nickel, thus driving increased demand. Non-EV demand such as stationary electronics/storage, non-battery, and flexible non-battery also saw a steady increase, but not as rapid as the EV sector. However, the gap between demand and production capacity is likely to widen around 2025, driven particularly by the rapid increase in EV demand, resulting in a potential nickel supply shortage.

However, a different fact is seen in the market. The spike in the 2024 surplus signifies the global nickel dynamics, especially caused by the developments in Indonesia. According to the International Nickel Study Group (INSG), the surplus is expected to widen to 239,000 metric tons by 2024, up from 223,000 tons in the previous year. This increase in surplus was largely due to increased production of low-grade “Class 2” nickel in Indonesia, particularly in the form of NPI.



Source : International Nickel Study Group 05/10/2023

Figure 7. INSG projection towards the nickel surplus in 2024²³

In the global nickel market, Indonesia plays an important role, especially with its NPI production. It is encouraged by the abundant nickel resources in Indonesia and the government policy that boosted the domestic processing of nickel. Sumitomo Metal Mining, Japan’s largest nickel smelter, predicts that global demand for nickel in 2024 will increase by 8.5% from the previous year to 3.35 million tons, with supply rising by 8.9% to 3.39 million tons. This increase in demand is expected due to the growth of EV markets in China and Europe, as well as the gradual expansion in the stainless-steel manufacturing sector.

²³ <https://www.reuters.com/markets/commodities/insgs-latest-forecasts-cold-comfort-nickel-producers-2024-04-25/>

Meanwhile, nickel prices have been on the decline, currently trading lower than the beginning of the year. This is due to the nickel class-2 surplus, despite the tight market for nickel class-1. This price decline can affect the viability of new mining projects, as already seen with various companies slowing down or suspending their operations.

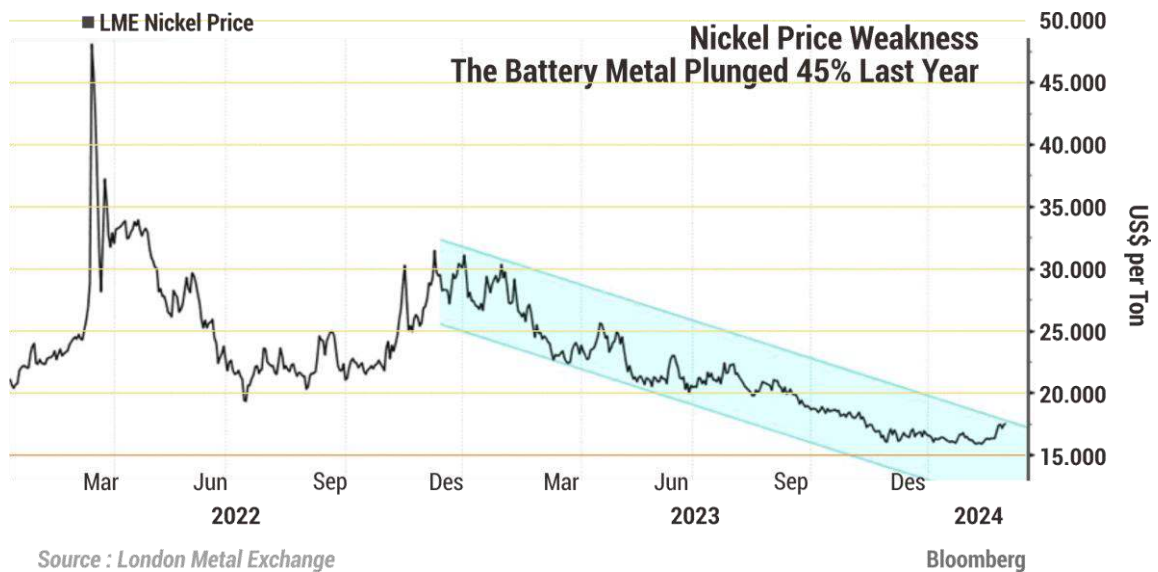


Figure 8. Nickel price drop trend²⁴

Overall, the global nickel market in 2024 faces a complex situation with supply increase, especially from Indonesia, and growing demand driven by the EV sector and stainless-steel production. However, surplus and price dynamics pose challenges to the industry, affecting investment decisions and the viability of new mining projects.

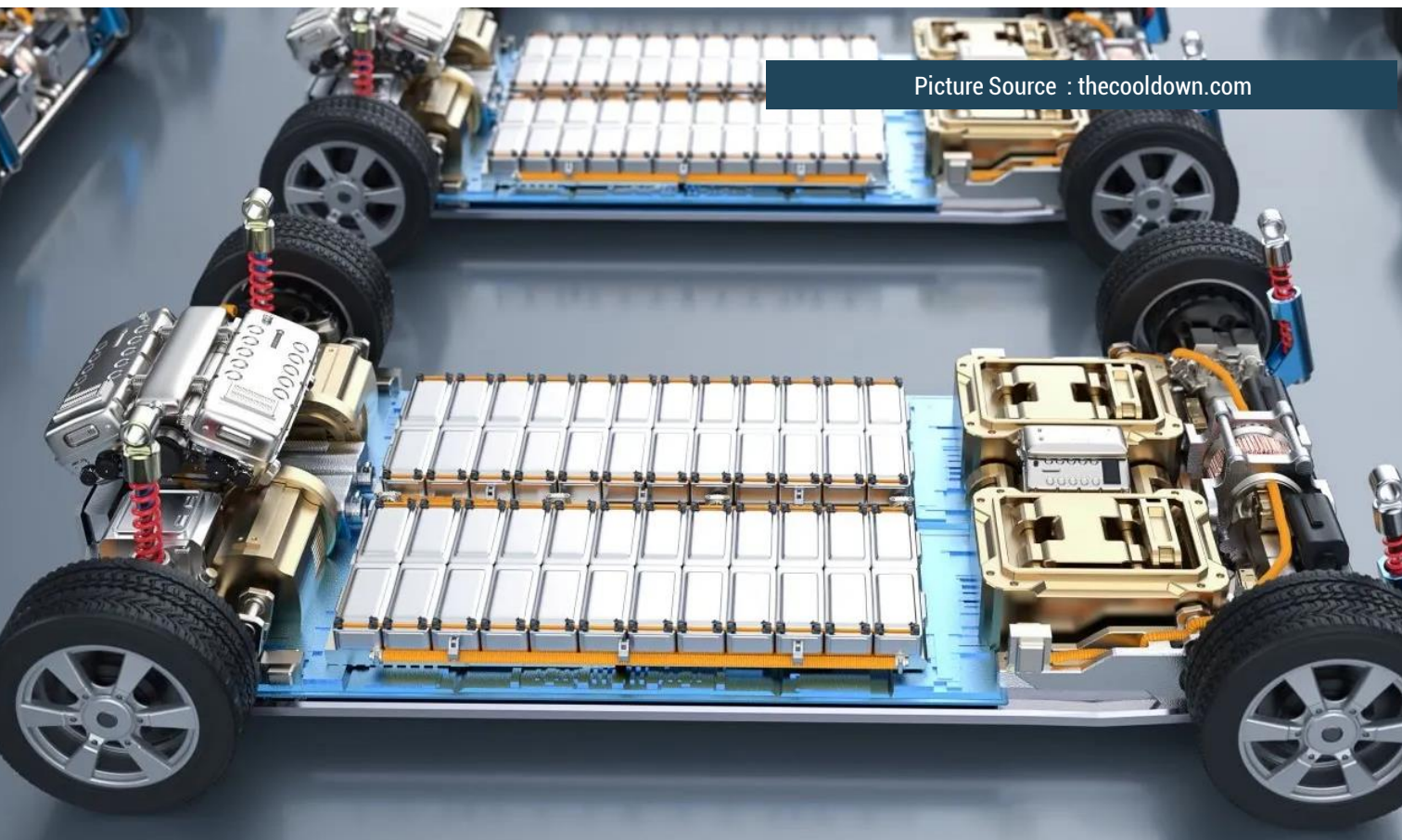
²⁴ <https://www.bloomberg.com/news/articles/2024-02-28/indonesia-sees-its-nickel-processing-capacity-keeping-prices-low>



Picture Source : pxhere.com

3. Nickel Supply Chain from Indonesia to Tesla, BYD, & Volkswagen

Of the many global battery supply companies that supply to many EV manufacturers, the top ten companies are Contemporary Amperex Technology (CATL), LG Energy Solution, Panasonic, Samsung SDI, BYD Co., SK Innovation, China Aviation Lithium Battery (CALB), Gotion High-Tech, Automotive Energy Supply Corp. (AESC), and Ruipu Energy Co. (REPT). Tesla, BYD, and VW as the three biggest EV manufacturers also use battery supplies from some of those top battery supply companies. Of the ten battery suppliers, some of them have entered a contract in 2021 with the three biggest manufacturers of electric vehicles (Tesla, BYD, VW).



Picture Source : thecooldown.com

Table 1. Relation between battery supplier companies and EV manufacturers²⁵

No	Battery Supplier	EV manufacturers (already in contract)
1	Contemporary Amperex Technology (CATL)	• BMW • Dongfeng Motor Corp • Honda • SAIC Motor Corp • Stellantis • Tesla • VW Group • Volvo Car Group
2	LG Energy Solution	• General Motors • Groupe Renault • Stellantis • Tesla • Volvo • VW Group
3	Panasonic	• Tesla • Toyota
4	Samsung SDI	• BMW • Ford • Stellantis • VW Group
5	BYD Co.	• BYD • Ford
6	SK Innovation	• Daimler • Ford • Hyundai • Kia
7	China Aviation Lithium Battery (CALB)	• GAC Motor • Zhejiang Geely • Holding Group Co
8	Gotion High-Tech	• Chery Automobile Co • SAIC • VW Group
9	Automotive Energy Supply Corp (AESC)	• Groupe Renault • Nissan
10	Ruipu Energy Co. (REPT)	• Dongfeng • Yudo Auto

²⁵ <https://spectrum.ieee.org/the-top-10-ev-battery-makers>

Contemporary Amperex Technology (CATL), LG Energy Solution, Panasonic, Samsung SDI, BYD Co., and Gotion High-Tech, which supply batteries for Tesla, BYD, and VW, obtains their nickel supplies from Indonesia through a nickel industrial company, Huayou Indonesia, which originates from China and is part of Zhejiang Huayou Cobalt. This can be seen on Huayou Indonesia's website, as shown in Figure 8 below. To process nickel, Huayou Indonesia uses a captive coal power plant (PLTU) to maintain stability in electricity generation. The related explanation can be seen in Section 5.



Figure 9. Battery companies supplied by Huayou Indonesia²⁶

²⁶ <https://www.huayouindonesia.com/id/about-us/global>



Picture Source : Sergii-Chernov / shutterstock.com

4. Relevance of Environmental, Social, and Governance (ESG) Assessment and PROPER Rating in the Nickel Industry for EV Batteries

ESG is an international standard for evaluating business performance with the purpose of assessing the sustainability risks of a company, by examining the company's ability to manage risks and respond to climate change, epidemics, social issues, climate policies, and other issues.

With increasing demand for critical minerals for low-carbon technologies, including nickel, the mining and battery production industry faces both opportunities and challenges to improve their ESG practices. This transformation is not only important to meet global climate targets, but also to maintain the confidence of investors and other stakeholders.

ESG has a very important relevance in the industry of mining and battery production, especially in Indonesia, a significant nickel producer for electric vehicle batteries. Here are some key aspects of ESG relevance in the industry:

1. **Environmental:** Mining has significant impacts on the environment, including greenhouse gas emissions and waste management. In Indonesia, with a focus on nickel mining, there is pressure to adopt more environmentally friendly practices to minimize negative impacts on ecosystems and to comply with global standards. The mining industry also contributes to deforestation and biodiversity loss, which requires effective mitigation efforts.
2. **Social:** The mining industry affects local communities, including labor, health, and safety issues. In Indonesia, there is a need to ensure that the rights and welfare of local communities and workers are respected, as well as providing equitable economic benefits from extracted resources.
3. **Governance:** Governance aspects involve regulatory compliance, transparency, and corruption prevention. This includes responsible management of natural resources and fairness in profit distribution. In Indonesia, these challenges are related to effective regulation and the prevention of unethical business practices.

From a global perspective, ESG trends in the mining and metals industry emphasize the importance of considering impacts on climate and society, and adopting ESG-driven business models. Meanwhile, specifically Indonesia as the biggest nickel producer in the world, the country faces unique challenges and opportunities in implementing ESG standards in the mining and battery production industries. The adoption of ESG in the mining and battery production industry is not only important for environmental sustainability but also for social justice and good governance, all of which are highly relevant in the global context and particularly in Indonesia.

The ESG framework is used as a principle that underpins responsible business practices by helping organizations manage environmental, social, and governance aspects. This framework provides a structured reporting scheme that ensures coherence and consistency in ESG reporting.

ESG reporting acts as an avenue for organizations to communicate ESG issues to stakeholders, especially investors. By referring to the framework, organizations can ensure that their ESG initiatives can deliver measurable, actionable, and credible results.

ESG standard is more precise and detailed with specific guidelines related to sustainability reporting. ESG standard covers not only the issues handled by the company, but also how the company manages the issues. Different standards and methods of ESG assessment are applied to different industries. Aspects in ESG standards cover:

- Specific metrics and indicators that must be measured by the company. For instance, emission rate, percentage of women within the workforce, etc.
- Methodology: ESG standards describe the methodology and calculation required in measuring the company's ESG. This ensures the consistency and comparability of reporting across organizations.
- Reporting format: ESG standard guides the format and structure that the organizations are required to use in presenting their ESG performance.

In Indonesia, the Financial Services Authority (*OJK*) through Financial Services Authority Regulation (*POJK*) Number 51/2017 on the Implementation of Sustainable Finance for Financial Services Institutions, Issuers, and Public Companies requires companies listed on the Indonesia Stock Exchange (*IDX*) to be open and transparent about ESG reports. Companies not listed on the *IDX* are not required to disclose their ESG reports to the public. *IDX* collaborates with Morningstar Sustainalytics to perform ESG assessments. ESG scores have five categories in the range of 0-40. An illustration of the score categorization can be seen in Table 2 below.

Table 2. ESG Score and its categories²⁷

No	ESG Score	Category	Description
1.	0-10	Negligible	Considered to have negligible ESG risk
2.	10-20	Low	Considered to have low ESG risk
3.	20-30	Medium	Considered to have moderate ESG risk
4.	30-40	High	Considered to have a high ESG risks
5.	>40	Severe	Considered to have a severe ESG risk

The dimensions of ESG assessment included in the *OJK* assessment can be seen in Table 3.

Table 3. ESG assessment dimensions by *OJK*

Environmental	Social	Governance
Use of environment-friendly energy	Selection of suppliers with ESG policy and practice	Use of accounting methods that comply with required standards
Waste processing to prevent pollutants	Organizational involvement in community development either in the form of profits percentage or employee volunteering for the community	Assurance that all related parties are provided with opportunities to participate in voting for decisions on issues of importance to the country
Participation in the conservation of irreplaceable natural resources	Assurance of a healthy and safe work environment for workers	Assurance that there will be no political contribution to obtain preferential treatment from the contribution recipient
Reasonable, non-arbitrary treatment of animals	Assurance to consider the stakeholders' inputs and wishes for the organization	Assurance of non-involvement in illegal activities
The implementation of an effective risk management system in environmental risk management		

²⁷ <https://www.idx.co.id/en/listed-companies/esg-score-explanation/>

At the national level, apart from the non-mandatory ESG standard for all levels of industry, there is another standard being used to assess environmental management by businesses, that is PROPER (*Program Penilaian Peringkat Kinerja Perusahaan dalam Pengelolaan Lingkungan Hidup* or Company Performance Rating Program in Environmental Management). PROPER is required by the Ministry of Environment and Forestry (*Kementerian Lingkungan Hidup dan Kehutanan/KLHK*) for every company with activities that cause a significant impact on the environment, regardless of the company being listed or not listed on the IDX.

The basic principle of PROPER implementation is to encourage company governance in environmental management through reputation/image incentive instruments for companies with good environmental management performance and reputation/image disincentive instruments for companies with poor environmental management performance. PROPER assesses the holistic performance of environmental management (of wastewater, gas emission, and toxic & hazardous waste). PROPER does not cover the social and governance aspects.

Business performance rating and or activity given consists of:²⁸

- a. Gold is for businesses and/or activities that have consistently demonstrated environmental excellence in the production or service process, conduct ethical business, and are responsible to the community.
- b. Green is for businesses and/or activities that have carried out environmental management more than required in the regulations (beyond compliance) through the implementation of an environmental management system, efficient utilization of resources, and good social responsibility efforts.
- c. Blue is for businesses and/or activities that have carried out the required environmental management efforts under the applicable laws and regulations.
- d. Red is an environmental management effort that has not been carried out in accordance with the requirements as stipulated in the laws and regulations.
- e. Black is for businesses and/or activities that deliberately commit acts or commit omissions that result in environmental pollution or damage as well as violations of applicable laws and regulations or do not implement administrative sanctions.

²⁸ Ministry of Environment and Forestry Regulation No. 1 of 2021 on Company Performance Rating Program in Environmental Management

5. Identification of ESG & PROPER Assessment of Nickel Production Company for EV Battery

Nickel processing companies in Indonesia are concentrated on Sulawesi Island and the Maluku Islands, spread across Bantaeng, Morowali, North Morowali, Konawe, Weda, and Obi Island. This processing company's activities certainly impact the surrounding social-environmental, which is outlined in the Environmental Impact Assessment (AMDAL) before the company's activities begin. After the processing activities are running, the impact on the social-environmental is evaluated, either internally by the company itself or by the service of a third party. Both the ESG assessment and the PROPER award are good steps to analyze and evaluate the impacts of smelter companies for improvements towards sustainability.



Picture Source : AEER / Arfah

The majority of smelter companies in Indonesia are not yet listed on the Indonesia Stock Exchange, so there is no pressure to evaluate and publicize the environmental, social, and governance risks of their activities. In the following are companies with nickel smelting activities that are listed on the Indonesian Stock Exchange and their ESG scores.

Table 4. Companies with nickel management activities listed on the IDX and their ESG scores
(Source: BEI & Morningstar Sustainalytics)

No	Company Name	Exploration/Mining/Nickel Processing Activities	ESG Score	Description
1.	PT Trimegah Bangun Persada Tbk	PT Gane Permai Sentosa, PT Gane Tambang Sentosa, PT Halmahera Jaya Feronikel, PT Jikodolong Megah Pertiwi, PT Megah Surya Pertiwi, PT Obi Anugerah Mineral	41.3	Severe impact on the environment and society with serious business risks.
2.	PT Vale Indonesia Tbk	PT Vale Indonesia Tbk	30.5	High impact on the environment and society with high business risk.
3.	PT Aneka Tambang Tbk	PT Gag Nikel, PT Nusa Karya Arindo, PT Sumberdaya Arindo	40.5	Severe impact on the environment and society with serious business risks.
4.	PT Central Omega Resources Tbk	PT Afit Lintas Jaya, PT Bumi Konawe Abadi, PT COR Industri Indonesia, PT Itamatra Nusantara, PT Mega Buana Resources, PT Mulia Pacific Resources	N/A	N/A
5.	PT Ifishdeco Tbk	PT Berkat Nikel Indonesia, PT Berkat Nikel Kolaka, PT Bintang Smelter Indonesia, PT Patrindo Jaya Makmur, PT Sarana Bumi Minerindo	N/A	N/A
Description: N/A: data not available				

Aside of PT Vale Indonesia, the connection of the four companies in Table 4 with nickel battery production in Indonesia is yet to be confirmed. In 2023, PT Vale Indonesia signed a cooperation agreement for the construction of a nickel processing facility with HPAL technology in Maili, East Luwu to produce MHP that can be used in battery manufacturing. Unfortunately, the nickel smelters for battery manufacturers in Indonesia (Huayue NC, QMB NEM, and Halmahera PL) are not even listed on the IDX, that they do not have the obligation to report their business practice in accordance with the framework and standards of the ESG. This indicates a gap between ESG demands and the smelting practices of EV battery manufacturers. Nickel smelter companies may avoid ESG requirements by avoiding listing their companies on the IDX.

Another challenge is the lack of transparency on companies' ESG assessments in the form of sustainability reports. This study discovered nickel production companies publicly listed on the IDX not yet transparent and open on the risks of their activity to the environment, social, and governance. The public does not even have information access to the scores of each metric in the ESG assessment. In addition, what the public experiences, especially the local community, are often not congruent to the company report contents. Some of them are: the eviction of pepper farmers by PT Vale Indonesia in South Sulawesi,²⁹ massive numbers of ARI in children in Morowali after PT IMIP began operation,³⁰ environmental pollution by PT Ifishdeco³¹ mine, heavy metals pollution in Obi Island³², a decline in the quality of life of local communities.³³

Picture Source : AEER / Arfah



²⁹ <https://www.betahita.id/news/detail/8458/walhi-sulsel-desak-pt-vale-hentikan-operasi-tambang.html?v=1711889046>

³⁰ <https://www.detik.com/sulsel/berita/d-6978060/warga-morowali-curhat-rumah-rusak-anak-ispa-gegara-debu-batu-bara-pt-imip>

³¹ Aldiansyah, S. & Nursalam, L.O. (2019). The impact of PT Ifishdeco Nickel Mining to the Environmental Conditions in Roraya Village, Tinanggea Sub-district, South Konawe District. Jurnal Penelitian Pendidikan Geografi Volume 4 Nomor 1

³² <https://www.mongabay.co.id/2023/04/16/mereka-suarakan-kerusakan-pulau-obi-dampak-industri-nikel/>

³³ Aksi Ekologi & Emansipasi Rakyat. (2020). Rangkaian Pasok Nikel Baterai dari Indonesia dan Persoalan Sosial Ekologi. <https://www.aeer.or.id/rangkaian-pasok-nikel-baterai-di-indonesia-dan-persoalan-sosial-ekologi/>

In addition to ESG reporting, companies' commitment to social-environmental performance evaluation can also be identified from the PROPER awards they receive. In the IMIP area, Huayou has a joint venture smelter managed by PT Huayue Nickel Cobalt. The following are nickel processing companies in the IMIP area and their PROPER awards.

Table 5. PROPER rating of Tenant in PT IMIP
(Source: Ministry of Environment and Forestry)

No	Company Name	PROPER Award	
		2022	2023
1.	PT Indonesia Morowali Industrial Park	Red Rating	Suspended Rating
2.	PT Bintangdelapan Mineral	Blue Rating	Blue Rating
3.	PT Sulawesi Mining Investment	Blue Rating	Red Rating
4.	PT Indonesia Guang Ching Nickel and Stainless Steel Industry	Blue Rating	Red Rating
5.	PT Indonesia Ruipu Nickel and Chrome Alloy	-	Red Rating
6.	PT Indonesia Tsingshan Stainless Steel	Red Rating	Red rating
7.	PT Dexin Steel Indonesia	-	Blue Rating
8.	PT Hengjaya Mineralindo	Green Rating	Green Rating
9.	PT Ranger Nickel Industry	-	Red Rating
10.	PT Huayue Nickel Cobalt	-	Suspended Rating
11.	PT QMB New Energy Materials	-	Red Rating
12.	PT Lestari Smelter Indonesia	-	Red Rating
13.	Bintang Delapan Terminal	N/A	N/A
14.	PT Indonesia Puqing Recycling Technology	N/A	N/A
15.	Fajar Metal Industry	N/A	N/A
Description: N/A: data not available			

Other than the IMIP area, there is also the Indonesia Weda Bay Industrial Park (IWIP) area that is the center of nickel production in Central Halmahera. One of the IWIP stock owners is Huayou. PROPER ratings of tenant companies in IWIP are as follows:

Table 6. PROPER rating of Tenant in PT IWIP
(Source: Ministry of Environment and Forestry)

No	Company Name	PROPER Award	
		2022	2023
1.	PT Weda Bay Nickel	Red Rating	Suspended Rating
2.	PT Yashi Indonesia Investment	N/A	N/A
3.	PT Youshan Nickel Indonesia	N/A	N/A
Description: N/A: data not available			

From Table 5 and Table 6 above, it could be seen that for national evaluation done by the KLHK, nickel smelter companies have not seriously reduced the negative impact of their activities on the environment. Companies with a PROPER red rating receive administrative sanction in the form of instruction to improve and the obligation to attend training for at least three (3) months after the PROPER rating is determined³⁴. However, the obtained data shows that companies with red ratings or suspended ratings in the last two years have not undergone any improvement to become better. It can be concluded that there is no improvement in environmental management by the company. EV producers need to consider the PROPER rating of the companies that supply batteries and their components for their production and terminate the supply contract from such companies with a low rating.

³⁴ Ministry of Environment and Forestry Regulation No. 1 of 2021 on Company Performance Rating Program in Environmental Management





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